Ericsson PE Node Level Routing Protocol Flap SOP (Minor Alarm 2486)

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_2486

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps required to mitigate a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on Ericsson PE Node WHI-PE-14 located at Whitefield East POP. This failure domain occurs on this specific role and vendor equipment due to the inherent complexities of routing protocol interactions, which can lead to transient oscillations in the control plane. The PE node's role in interconnecting multiple network segments, utilizing BGP, OSPF, and ISIS protocols, makes it susceptible to routing protocol flapping. Furthermore, the use of Ericsson's proprietary CLI dialect and hardware-specific implementations of these protocols add an additional layer of complexity, necessitating a thorough and vendor-specific approach to diagnosis and resolution. The underlying protocol mechanics, including the use of TCP-based session establishment for BGP and OSPF, and the reliance on LSP flooding for ISIS, contribute to the occurrence of routing flapping. Additionally, hardware-related factors, such as port or interface resets, can also trigger routing protocol flapping.

Pre-requisite Checks

- Verify that the Whitefield East POP site is accessible and the remote hands are dispatched to the location

- Check the status of redundant links and ensure that they are operational

- Review the node's configuration and ensure that it is consistent with the vendor's recommendations

- Verify that the node's software and firmware are up-to-date and running the latest versions

Potential Root Causes

Cause	Probability
BGP route churn due to incorrect routing policy configuration	High
OSPF adjacency loss due to network partitioning	Medium
ISIS LSP flooding due to hardware-related issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp neighbor *, Expected Output: Neighbor AS InQ OutQ Up/Down State Last RxRqmt TxRqmt, Description: This command is used to verify the BGP neighbor relationships and identify any issues with route exchange

Command: show ospf neighbor *, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command is used to
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verify the OSPF neighbor relationships and identify any issues with
adjacency formation

Command: show isis database *, Expected Output: IS-IS Level-1
Database, Description: This command is used to verify the ISIS LSP
database and identify any issues with LSP flooding
```

Resolution Steps

Step 1: Step: Verify and Correct BGP Routing Policy Configuration, Command: show bgp policy *, Description: Verify the BGP routing policy configuration and correct any incorrect policies that may be causing route churn

Step 2: Step: Verify and Correct OSPF Neighbor Relationships, Command: show ospf neighbor *, Description: Verify the OSPF neighbor relationships and correct any issues with adjacency formation

Step 3: Step: Verify and Correct ISIS LSP Flooding, Command: show isis database *, Description: Verify the ISIS LSP database and correct any issues with LSP flooding

Post-Check Verification

```
Command: show bgp neighbor *, Expected Output: Neighbor AS InQ OutQ
Up/Down State Last RxRqmt TxRqmt, Description: Verify that the BGP
neighbor relationships are stable and that route exchange is occurring
correctly

Command: show ospf neighbor *, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify that the OSPF
neighbor relationships are stable and that adjacency formation is
occurring correctly

Command: show isis database *, Expected Output: IS-IS Level-1
Database, Description: Verify that the ISIS LSP database is stable and
that LSP flooding is occurring correctly
```

Roll-back Procedures

- Step: Restore Previous BGP Routing Policy Configuration, Command: show bgp policy * | include previous-config, Description: Restore the previous BGP routing policy configuration if the fix fails

- Step: Restore Previous OSPF Neighbor Relationships, Command: show ospf neighbor * | include previous-config, Description: Restore the previous OSPF neighbor relationships if the fix fails

- Step: Restore Previous ISIS LSP Flooding Configuration, Command: show isis database * | include previous-config, Description: Restore the previous ISIS LSP flooding configuration if the fix fails